SMART ATTENDANCE SYSTEM

Project guide | Hinglish edition

Workflow, technologies, database, security, setup and testing

WHAT THIS PROJECT DOES

Teacher session banata hai; student session QR scan karke verified check-in karta hai.

Is guide mein

- End-to-end teacher aur student attendance workflow
- React, Express, Node.js, MySQL/JSON aur QR technologies ka role
- Data model, API routes, authentication aur duplicate prevention
- Local run, mobile/LAN testing, deployment aur verification

CURRENT IMPLEMENTATION

React teacher portal + Express API; local JSON mode and MySQL production mode.

Prepared from the existing project files | 23 October 2026

Project documentation - no credentials or private secrets are included.

1. System kaise kaam karta hai

Teacher portal se student phone tak attendance ka poora flow

Teacher workflow

- Teacher pehli baar Create teacher account se ek hi teacher account banata hai; baad mein login karta hai.
- Teacher portal mein registered students wale class ko select karta hai aur subject type karta hai.
- Start attendance session par server session ID, start time, expiry aur status banata hai. Default expiry 10 minutes hai; allowed range 1 se 120 minutes.
- Dashboard session-specific signed QR code dikhata hai. Teacher session ko expiry se pehle end bhi kar sakta hai.

Student check-in workflow

- Student apne phone ke Camera app ya Google Lens se teacher screen par dikh raha QR scan karta hai. Student ko teacher portal mein scan nahi karna hota.
- QR mobile-friendly website ke /student-scan page ko kholta hai. Page token ko server se validate karta hai.
- Student registered full name aur college roll number/student ID enter karta hai. Naam/ID class ke registered student se milne chahiye.
- Server active session, expiry, signed token, student identity aur duplicate attendance check karta hai.
- Sahi submission par session, student, class, subject, date, timestamp aur status ke saath present record save hota hai. Student ko success/error message milta hai.

Teacher results

- Dashboard active session ka present count aur checked-in students refresh karke dikhata hai.
- Reports mein session history, har session ke attendees, student-wise summary aur per-session CSV export milta hai.

Important

Yahan attendance teacher ke QR ko student ke phone se scan karke lagti hai. Student form mein submit karna identity verification ke baad check-in action hai; manual teacher-side camera scanner istemal nahi hota.

2. Technology stack

Kaunsi technology hai, aur project mein kis kaam aati hai

Frontend / browser

- React 19: component-based user interface; teacher login, dashboard, student scan page, settings aur reports.
- Create React App / react-scripts 5: development server, tests aur optimized production build.
- React Router DOM 7: /login, protected teacher portal, /reports, /settings aur public /student-scan routes.
- JavaScript + CSS: form state, responsive layout, mobile-friendly screens aur API interactions.
- qrcode 1.5: teacher portal mein signed attendance URL ko high-contrast QR image mein render karta hai.

Backend / server

- Node.js: JavaScript runtime; API process, cryptography aur file/database access.
- Express 5: REST API routes, request parsing, CORS policy, authentication middleware aur error responses.
- cors: frontend origin ko allow-list ke zariye API se baat karne deta hai.
- dotenv: local .env mein rakhe server settings load karta hai; secrets source code mein nahi rakhe jate.
- Node crypto: scrypt password hashing, random salt, HMAC-SHA256 teacher/QR signing, UUID session IDs aur constant-time comparisons.

Storage, deployment, tests

- JSON file: local development aur simple single-process testing ka store (server/data.json ya DATA_FILE path).
- MySQL + mysql2/promise: persistent deployment database, SQL queries, transactions, row locks aur unique constraints.
- Vercel config: React static build aur SPA route rewrites. Render config: Node MySQL API service.
- Node built-in test runner + React Testing Library/Jest: API integration aur frontend tests.

Do alag modes

npm run server JSON store chalata hai. npm run server:mysql MySQL mode chalata hai aur MYSQL_* credentials/schema maangta hai. Local JSON aur production MySQL ka data ek hi store nahi hota.

3. Database, identity aur security

Attendance kis data se banti hai aur server kaise validation karta hai

Main database entities

- students: id, full name, unique roll number/student ID aur class. Student registration teacher-protected API se hoti hai.
- teacher_accounts: one-time signup se ek teacher; username, salted password hash aur created time. Plain password save nahi hota.
- attendance_sessions: random session ID, teacher, class, subject, date, started/expires/ended timestamps aur active/ended/expired status.
- attendance: student ID/name/roll number, nullable legacy session ID, class, subject, date, status, source aur recorded timestamp.

Trust aur duplicate rules

- Name aur ID ki matching backend par registered student + session class se hoti hai; frontend-only checks par bharosa nahi.
- Signed QR token session ID aur expiry ko AUTH_SECRET se HMAC-SHA256 sign karta hai; invalid, ended ya expired QR reject hota hai.
- Teacher token signed hota hai aur 8 ghante valid rehta hai. Protected dashboard APIs Authorization bearer token maangti hain.
- Login/first-registration attempts par in-process rate limit: 15 minute mein per client IP 10 attempts tak.
- MySQL attendance insertion transaction mein hoti hai; UNIQUE(session_id, student_id) database constraint ek student ki same session mein duplicate entry rokta hai.
- JSON mode duplicate check karta hai, lekin multi-process/concurrent production writes ke liye MySQL jaisi transactional guarantee nahi deta.

Secrets aur network

- AUTH_SECRET kam-se-kam 32 characters ka random secret ho; .env ko share ya commit nahi karna. .env.example sirf template hai.
- CORS_ORIGIN sirf configured frontend origin ko allow kare. Deployed app ke liye HTTPS frontend URL PUBLIC_APP_URL mein ho.
- Teacher access token browser sessionStorage mein rehta hai; logout par remove hota hai.

Privacy note

QR ek temporary attendance link hai. Student submission mein naam aur ID collect hote hain; QR ko class ke bahar share na karein aur attendance exports ko secure rakhein.

4. Screens aur API map

Frontend pages aur backend endpoints ki zimmedari

Frontend screens

- Login: account status check, one-time first teacher signup, phir teacher sign in.
- Teacher home / attendance sessions: class + subject choose, start/end session, QR, present count aur live student list.
- Student scan page: public signed-token validation; full name + student ID form; success/error result.
- Reports: session history, attendee table, per-session CSV export, student-wise attendance percentage aur low-attendance alerts.
- Settings: school name, class options aur location fields. Attendance radius/location enforcement testing ke liye disabled hai.

Important API routes

- GET /api/health - service/database health.
- GET /api/auth/status, POST /api/auth/register, POST /api/auth/login - first signup state aur teacher authentication.
- GET /api/auth/me, GET /api/bootstrap, GET /api/attendance - teacher-only identity aur dashboard data.
- POST /api/sessions, GET /api/sessions, POST /api/sessions/:id/end - session lifecycle (teacher-only).
- GET /api/student-sessions/:token - public QR/session validation; only session metadata returned.
- POST /api/student-sessions/:token/attendance - public check-in request, but backend token/identity/session validation required.
- POST /api/students, DELETE /api/students/:id, PUT /api/settings - teacher-protected management APIs.

Retired endpoint

POST /api/attendance/scan jaan-boojhkar 410 Gone deta hai. Purane direct/manual attendance flow se record nahi banta.

Frontend-to-backend request

Development mein React ka /api proxy localhost:4000 par request bhejta hai. Deployment par frontend build setting REACT_APP_API_URL ko API ke /api base URL par set karti hai. API error aane par client explicit error message dikhata hai; fake success fallback nahi hai.

5. Project ko run aur deploy kaise karein

Local development, phone par test aur deployment setup

Local development

```
# Terminal 1: API (JSON store)
npm run server

# Terminal 2: React frontend
npm run dev
```

Frontend aam taur par http://localhost:3000/login par aur API http://localhost:4000 par hoti hai. Pehli baar page par ek teacher username aur kam-se-kam 12-character password create karein.

- .env mein random AUTH_SECRET (minimum 32 characters), SESSION_TTL_MINUTES, PUBLIC_APP_URL aur CORS_ORIGIN set karein. Current default expiry 10 minutes hai.
- Environment template: .env.example. Password/secret ko PDF, git ya public deployment text mein kabhi na likhein.

Phone se same Wi-Fi par test

- PC aur phone ek hi Wi-Fi par hon. PC ka LAN IPv4 address maloom karein; phone par localhost nahi, PC ka LAN IP use hota hai.
- React ko network interface par bind karein: PowerShell mein $env:HOST="0.0.0.0"; npm run dev.
- PUBLIC_APP_URL ko http://PC-LAN-IP:3000 aur CORS_ORIGIN ko exact same origin banakar API restart karein. Naya QR generate karein.
- QR expired/ended ho to naya attendance session start karein. Local HTTP camera site ke liye QR scan built-in Camera/Google Lens se hota hai; QR mein link phone ke browser mein khulta hai.
- Phone page tak nahi pahunchta to Wi-Fi client isolation, Windows firewall, VPN ya guest Wi-Fi check karein. Sirf trusted private LAN par ports 3000/4000 allow karein.

Production deployment

- Frontend ko static React build ke roop mein deploy karein; Vercel config SPA routes ko index.html par rewrite karti hai.
- API: npm run server:mysql; Render config isi command aur /api/health health-check ko use karti hai.
- MySQL server par server/schema.sql (fresh DB) ya existing DB ke liye backup ke baad one-time server/migrations/001_attendance_sessions.sql.
- Agar migration 001 pehle hi apply ho chuki hai to teacher account table ke liye 002_teacher_accounts.sql apply karein. Migration 001 ko repeat na karein.
- Deployment secrets: MYSQL_HOST, MYSQL_PORT, MYSQL_USER, MYSQL_PASSWORD, MYSQL_DATABASE, AUTH_SECRET, PUBLIC_APP_URL, CORS_ORIGIN, SESSION_TTL_MINUTES; frontend build par REACT_APP_API_URL.

Important

localhost sirf usi device ko refer karta hai. Bahar ke mobile networks ke liye public HTTPS deployment chahiye; LAN IP internet se accessible nahi hota.

6. Project files, verification aur known limits

Codebase map aur guide banate waqt check ki gayi capabilities

Key folders/files

- src/App.js, src/context/AppContext.js: routes, teacher auth state aur shared application state.
- src/pages/Login, StudentScan, Reports, Home, Settings: teacher login, student check-in, reports aur portal screens.
- src/components/AttendanceForm: session controls, QR generation, attendees/count; Dashboard, StudentList, AttendanceList: supporting UI.
- src/services/apiService.js: frontend HTTP client, bearer token aur API error handling.
- server/server.js: local JSON API; server/mysqlServer.js: MySQL API; server/auth.js: password/token/QR security helpers.
- server/schema.sql and server/migrations/: database schema aur existing DB upgrades; README.md: run/deploy instructions.

Project checks (current codebase)

- npm run build: pass; ek existing ESLint warning AttendanceList.js mein unused students variable ka hai.
- npm test -- --watchAll=false: frontend auth test pass.
- npm run test:server: 3 API integration tests pass - first registration/login, invalid identity, attendance insert, duplicate, reports, session end/expiry.

Known limits / use with care

- Automated end-to-end tests JSON backend par chalaye gaye. MySQL database service available na hone se migration/real MySQL workflow live verify nahi hua.
- JSON mode local/single-process use ke liye hai; production multi-user concurrency ke liye MySQL use karein.
- System mein one teacher account hai; multiple teacher roles/accounts abhi feature nahi hain. Student identity registered name + ID se check hoti hai; student login nahi hai.
- Location/radius checking disabled hai; QR validation ko GPS/proximity verification na samjhein.
- Settings screen ka Clear All Data purane localStorage data ko clear karta hai; server JSON/MySQL records ko delete karne ka mechanism nahi samjha jana chahiye. Is control ko backend data wipe na manen.

Data safety

server/data.json aur server/data-bitthal.json ko project data files ke roop mein treat karein. Backups ke bina delete na karein. Secrets .env mein hain; woh PDF mein include nahi kiye gaye.